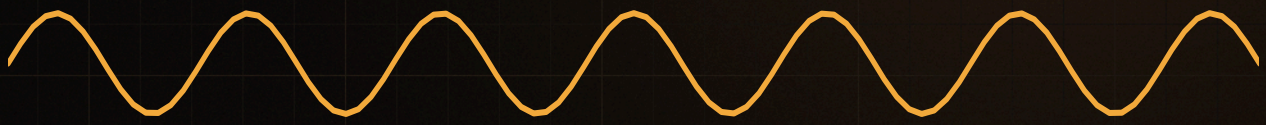
CHANNEL 01 · ACTIVE CH-01 · 433 MHz



SIGNAL LOCKED // 433.93 MHz · 4.02 Vpp · THD 1.1%

CLASSIFIED — PROBLEM STATEMENT CHANNEL

GENERAL INNOVATION & OPEN INNOVATION

IDEA STARTERS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29–30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 01 CH-01 · 433 MHz



SIGNAL LOCKED // 433.93 MHz · 4.02 Vpp · THD 1.1%

GENERAL INNOVATION & OPEN INNOVATION

Creative, non-traditional solutions for everyday challenges, accessibility and social welfare. Bridge hardware, software and human experience to build practical tools that don't fit into rigid silos.

— AREAS TO EXPLORE

Assistive Tech | Civic Hardware | Campus Utilities | Community IoT | Wearable Gear

— WHO SHOULD EXPLORE

Rapid Prototyping · Multidisciplinary Teams · Product Design & Human-Centric Hardware

> ***"Where unconventional thinking meets tangible impact — rethink everyday mechanics before the clock starts running."***

— IN THIS CHANNEL

12× Idea starters across energy, safety, mobility, utilities & wearables — or bring your own problem. OPEN

A · ABOUT THIS TRACK

The five domain channels contain detailed, structured problem statements. This Open Innovation track is deliberately lighter. It offers two routes: (1) pick one of the short, commonly seen idea starters in this document, such as a smart energy meter or a theft-protection system, and develop it in your own way, or (2) bring your own problem from your campus, home town, family business or daily life. Teams are free to choose either route; no domain restriction applies. No physical prototype or PCB is required in Round 1 — ideation, reasoning and simulation are what is judged.

B · WHAT EVERY TEAM SHOWS IN ROUND 1

- A clear problem: who faces it, how often, and what it costs them in money, time, safety or comfort.
- Existing solutions and their gaps: what is already available and why it is not enough (cost, complexity, accuracy, reach).
- Your idea: a system architecture or functional block diagram, and the hardware and software you would use.
- Proof of concept in simulation: a software simulation, analytical model or virtual demonstration with sample inputs and outputs.
- Feasibility: estimated cost, scalability and possible real-world applications.

C · SUGGESTED EVALUATION CRITERIA

Problem relevance	A real, well-understood problem with identified users and a measurable impact.	20%
Innovation	A fresh angle, a smart combination of ideas, or a clear improvement over existing solutions.	20%
Technical feasibility	Sensible component and technology choices; the idea would plausibly work outside the simulator.	20%
Simulation & demonstration	Working simulation or model with sample results that support the claims.	20%
Cost & scalability	Low-cost, practical, and able to grow from one unit to many.	10%
Presentation	Clarity of explanation, diagrams, and teamwork.	10%

Weights are suggested and may be adjusted by the organising committee.

— IDEA STARTERS — SHORT STARTING POINTS, DEFINE YOUR OWN SCOPE

OI-01

SMART ENERGY METER WITH USAGE ALERTS

Consumers see only a monthly bill, and conventional meters give no live data.

CONSIDER // Voltage, current, power, energy and power-factor measurement; live and historical usage; bill estimate; over-consumption and overload alerts.

BUILD // Energy-metering IC or CT + voltage sensor, microcontroller, Wi-Fi/BLE, display or app

OI-02

ELECTRICITY THEFT & METER TAMPERING DETECTION

Illegal tapping and meter bypass cause large losses to distribution utilities.

CONSIDER // Compare supply-side and consumer-side readings; detect bypass, reverse current and magnet or cover tampering; raise tamper alerts.

BUILD // CTs, Hall sensors, tilt/reed/cover switches, microcontroller, GSM/LoRa

OI-03

HOME & SHOP ANTI-THEFT SECURITY SYSTEM

Break-ins often go unnoticed until the owner returns.

CONSIDER // Door/window and motion sensing; arming and disarming; local siren; mobile alert; event log with time stamps.

BUILD // Reed switch, PIR sensor, buzzer, microcontroller, GSM/Wi-Fi

OI-04

VEHICLE THEFT PROTECTION & TRACKING

Two-wheelers and cars are stolen quickly, and recovery is difficult.

CONSIDER // Unauthorised-start or movement detection; engine immobilisation; location reporting; tamper and low-battery alerts.

BUILD // GPS module, accelerometer, relay, microcontroller, GSM

OI-05

SMART STREET LIGHT CONTROL

Street lights burn at full power all night, wasting energy, and faults go unreported.

CONSIDER // Ambient-light and motion based dimming; lamp-fault detection; energy saved against a fixed schedule.

BUILD // LDR, PIR/ultrasonic sensor, LED driver, microcontroller, LoRa/GSM

OI-06

AUTOMATIC PLANT WATERING & SOIL MONITORING

Plants and small farms are over- or under-watered, wasting water and yield.

CONSIDER // Soil-moisture and temperature sensing; pump control with limits; water-use log; low-tank warning.

BUILD // Soil-moisture sensor, pump and relay, microcontroller, optional solar

— IDEA STARTERS — SHORT STARTING POINTS, DEFINE YOUR OWN SCOPE

OI-07

SMART PARKING SLOT AVAILABILITY

Drivers waste time and fuel searching for parking.

CONSIDER // Slot occupancy detection; live count of free slots; entrance display; simple usage statistics.

BUILD // Ultrasonic or IR sensors, microcontroller, display, Wi-Fi

OI-08

CLASSROOM & LAB ENERGY SAVING VIA OCCUPANCY

Lights and fans stay on in empty rooms.

CONSIDER // Occupancy and light-level detection; automatic switch-off with delay; energy-saving estimate.

BUILD // PIR sensor, people counter, relay, microcontroller

OI-09

WATER TANK LEVEL MONITORING & OVERFLOW PREVENTION

Overhead tanks overflow or run dry unnoticed.

CONSIDER // Level measurement; automatic pump start/stop; overflow and dry-run protection; level display.

BUILD // Ultrasonic or float sensor, relay, microcontroller, optional app

OI-10

FIRE & SMOKE EARLY WARNING SYSTEM

Fires in homes, labs and godowns are detected late.

CONSIDER // Smoke, temperature and flame sensing; graded alarms; false-alarm filtering; remote notification.

BUILD // Smoke/temperature/flame sensors, microcontroller, buzzer, GSM

OI-11

EV CHARGING POINT MONITORING & BILLING

Small charging points lack metering, safety checks and fair billing.

CONSIDER // Energy metering per session; over-current and earth-leakage cut-off; user authentication; session billing.

BUILD // Metering IC, contactor, RFID, microcontroller, Wi-Fi

OI-12

WEARABLE PERSONAL SAFETY & EMERGENCY ALERT DEVICE

People in danger or with medical issues cannot always call for help.

CONSIDER // Panic button; fall detection; location sharing; automatic alert to contacts.

BUILD // Accelerometer, GPS, BLE/GSM, microcontroller, small battery

Other quick ideas, equally acceptable: battery-health monitor for e-bikes, smart door lock with access log, noise or air-quality monitor for classrooms, power-cut notifier, low-cost weather station, smart dustbin fill-level alert, baby-room monitor, smart attendance system.

D · CHOOSING & SHAPING YOUR OWN IDEA

A good open-innovation idea starts from something you have personally seen go wrong — on campus, in a hostel, a neighbourhood, a market or a relative's workplace. Then check it against these questions.

Is it real?	Can you name the people who face this problem and describe how they cope today?
Is it electronic?	Sensing, measuring, controlling, monitoring, communicating or saving energy should be central.
Can it be simulated?	You must be able to show it working in software, a model or a virtual demonstration within the hackathon time.
Affordable & safe?	Avoid ideas that need expensive equipment, hazardous voltages or regulated medical/defence approvals to be shown.
Original enough?	Explain clearly how your approach differs from products already on the market.

E · IDEA PROPOSAL TEMPLATE (SUBMIT WITH YOUR ROUND 1 ENTRY)

Title	A short, descriptive name for your idea.
Problem statement	Two to four sentences: the problem, who faces it and why it matters.
Existing solutions & gaps	What exists today and where it falls short.
Proposed solution	The idea in plain language, plus a block diagram of the system.
Key features	Three to six functions the system must perform.
Hardware & software	Sensors, controller, communication, power supply, tools and algorithms.
Simulation plan	What you will simulate, the tool you will use and the results you expect to show.
Impact & cost	Who benefits, estimated cost per unit and how it could scale.
Risks & limitations	What could go wrong, and what you would do next with more time.

F · TIPS FOR A STRONG SUBMISSION

- Keep the scope small enough to simulate well — a focused idea shown convincingly beats a broad idea described vaguely.
- Use numbers: assumed loads, thresholds, response times, costs and accuracy.
- Show at least one normal case and one fault or failure case in your simulation.
- Be honest about limitations; judges value realistic reasoning.